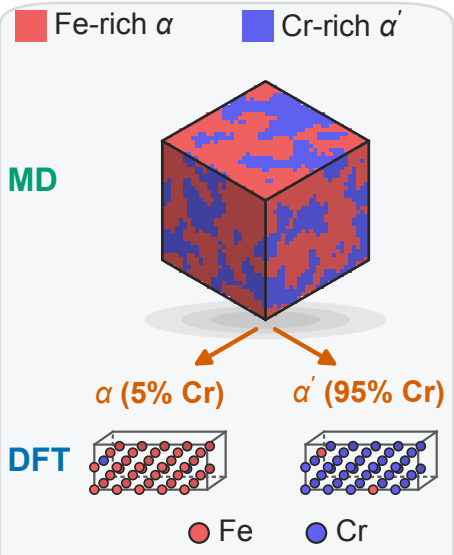
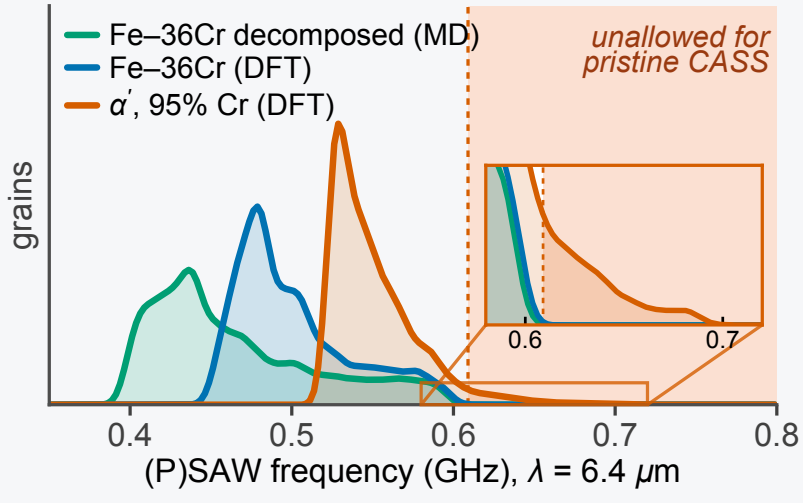
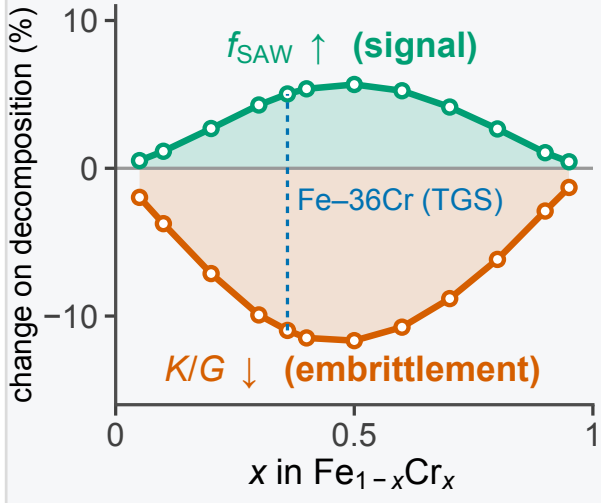



**Simulated spinodal decomposition of $Fe_{1-x}Cr_x$**




**DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite**




**Shear stiffening raises $f_{SAW}$ and lowers $K/G$: a signature of embrittlement**